

# Rebecca Morrison

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[CU Boulder](#)  
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## Positions

- Assistant Professor, Dept. of Computer Science, University of Colorado Boulder, August 2018 – present  
*Affiliate Faculty Member, Robotics Program*
- Postdoctoral Associate, Dept. of Aeronautics and Astronautics, Massachusetts Institute of Technology, 2016 – 2018. *Advisor: Prof. Youssef Marzouk*
- Intern, Sandia National Laboratories, Albuquerque, NM, 2012
- Student Researcher, International Research Experience for Students (IRES), Federal University of Ceará, Fortaleza, Brazil, 2008
- Undergraduate Researcher, REU, University of California Davis, 2007
- Undergraduate Researcher, Joint Science Department of the Claremont Colleges, 2005

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## Education

- Ph.D. in Computational Science, Engineering, and Mathematics, The University of Texas at Austin, 2016  
*Advisor: Prof. Robert Moser*
- M.S. in Computational and Applied Mathematics, The University of Texas at Austin, 2012
- B.A. in Physics, Scripps College, *Magna cum Laude*, Phi Beta Kappa, 2008

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## Honors & Awards

### *Awards & Fellowships*

- Exceptional Graduate Faculty Mentor Award, CU Boulder, 2025
- Johnson & Johnson Women in STEM2D (WiSTEM2D) Scholars Award, 2019
- Computational and Applied Mathematics Fellowship, UT Austin, 2009 – 2013
- Rudmose Department Fellowship, UT Austin, Physics Department, 2008
- Dart/Merritt Award for the Outstanding Student in Engineering or Physics, Joint Science Department of the Claremont Colleges, 2008

### *Paper, Presentation & Advisee Awards*

- Best Paper Award, IMAC 2023 — R. Bandy (advisee) and R. Morrison
- CERRA Student Recognition Award, ICASP14 2023 — T. Price-Broncucia (advisee)
- UQ Student Paper Competition Semi-Finalist, USNCCM 2023 — T. Price-Broncucia (advisee)
- Contest Winner, CRA Virtual Conference Lightning Talk, 2020
- Best Student Paper Award, World Congress on Engineering and Computer Science, 2011

## Grants & Contracts

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- Researcher in Residence Program, CU Museum of Natural History: *Resource distribution and the evolution of the division of labor* (PI). \$5,000, 2025.
- NSF Formal Methods in the Field: *Verified Probabilistic Programming for Hybrid Systems* (Co-PI; PI S. Sankaranarayanan, Co-PI G. Kaki). \$875,000 (share: \$291,667), 2024 – 2028.
- NASA Space Weather with Quantified Uncertainties: *Ensemble Learning for Accurate and Reliable Uncertainty Quantification* (Co-PI; PI E. Camporeale, Co-PI T. Berger). \$2,891,954 (share: \$469,299), 2020 – 2023.
- Johnson & Johnson Women in STEM2D Award: *Discovering Dynamic Structure from Data* (PI). \$150,000, 2019 – 2022.
- Association for Women in Mathematics/NSF Travel Grant, \$600, 2017

## Publications

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<sup>†</sup> (Co-)first author, otherwise listed first; \* PhD student, \*\* Master's student, \*\*\* Undergraduate student

### Journal Articles (Peer-reviewed)

14. **R. Morrison** and E. Basor. Exact mean and covariance formulas after diagonal transformations of a multivariate normal. *Journal of Multivariate Analysis*, page 105489, 2025.
13. T. Price-Broncucia\*, A. Baker, D. Hammerling, M. Duda, and **R. Morrison**. The ensemble consistency test: From CESM to MPAS and beyond. *Geoscientific Model Development*, 18:2349–2372, 2025.
12. S. Leventhal\*, S. Edie, **R. Morrison**, and C. Simpson. Origin of division of labor is decoupled from polymorphism in colonial animals. *PNAS Nexus*, 3(11):476, 2024. **Smithsonian National Museum of Natural History Early Career Award**.
11. E. Basor and **R. Morrison**<sup>†</sup>. Analytic solutions to nonlinear ODEs via spectral power series. *Linear Algebra and its Applications* 697:561–582, 2024.
10. R. Baptista\*<sup>†</sup>, **R. Morrison**<sup>†</sup>, O. Zahm, and Y. Marzouk. Learning non-Gaussian graphical models via Hessian scores and triangular transport. *Journal of Machine Learning Research* 25(85):1–46, 2024.
9. R. Bandy\* and **R. Morrison**. Stochastic model corrections for reduced Lotka-Volterra models exhibiting mutual, competitive, and predatory interactions. *Chaos: An Interdisciplinary Journal of Nonlinear Science* 34(1):013116, 2024.
8. **R. Morrison**, R. Baptista\*, and E. Basor. Diagonal nonlinear transformations preserve structure in covariance and precision matrices. *Journal of Multivariate Analysis*, page 190:104983, 2022.
7. M. Tosin, E. Dantas, A. Cunha, and **R. Morrison**. ARBO: Arbovirus Modeling and Uncertainty Quantification Toolbox. *Software Impacts* 12:100252, 2022.
6. **R. Morrison**. Exact reduction of the generalized Lotka-Volterra equations via integral and algebraic substitutions. *Computation*, 9(5):49, 2021.
5. **R. Morrison**. Data-driven corrections of partial Lotka-Volterra models. *Entropy* 22(11):1313, 2020.
4. **R. Morrison** and A. Cunha. Embedded model discrepancy: A case study of Zika modeling. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 30(5):051103, 2020.
3. **R. Morrison**, T. Oliver, and R. Moser. Representing model inadequacy: A stochastic operator approach. *SIAM/ASA Journal on Uncertainty Quantification* 6(2):457–496, 2018.
2. **R. Morrison**, C. Bryant, G. Terejanu, S. Prudhomme, and K. Miki. Data partition methodology for validation of predictive models. *Computers and Mathematics with Applications*, 66(10):2114–2125, 2013.

1. **R. Morrison**, A. Landsberg, and E. Friedman. Combinatorial games with a pass: A geometric approach. *Chaos: An Interdisciplinary Journal of Nonlinear Science* 21:043108, 2011.

#### Journal Articles — Accepted

1. T. Price-Broncucia\* and **R. Morrison**. Ultra-Fast unscented Kalman inversion for the calibration of expensive chaotic models. *SIAM Journal on Scientific Computing*.

#### Journal Articles — In Review

3. R. Bandy\*, E. Camporeale, A. Hu, T. Berger, and **R. Morrison**. Accurate and Reliable Uncertainty Estimates for Deterministic Predictions: Extensions to Under and Overpredictions. *SIAM Journal on Uncertainty Quantification*. arXiv:2604.08755.
2. R. Bandy\*, **R. Morrison**, E. Mussoni, and T. Portone. Hybrid physics-data enrichments to represent uncertainty: A case study of reduced gas-surface chemistry models for hypersonic flight. *International Journal of Uncertainty Quantification* arXiv:2509.08137.
1. U. Shah\*, M. Lladser, and **R. Morrison**. Conditional independence estimates for the generalized nonparamormal. *Journal of Machine Learning Research* arXiv:2508.11050.

#### Preprints / In Preparation

1. P. Marcy and R. Morrison. “Stochastic Inverse Problems” and changes-of-variables. arXiv:2211.15730

#### Peer-Reviewed Conference Papers

Acceptance rate and journal equivalence noted when applicable.

9. R. Bandy\*, R. Washington\*\*, **R. Morrison**, and T. Portone. Isolating and quantifying uncertainties in the vibration isolation round-robin challenge. In *Model Validation and Uncertainty Quantification, Vol. 3*, pages 71–85. River Publishers, 2025.
8. S. Liaw\*\*\*, **R. Morrison**, Y. Marzouk, and R. Baptista. Learning local neighborhoods of non-Gaussian graphical models: A measure transport approach. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39, No. 18, 2025. (Acceptance rate: 23%; journal equivalent) Extended version: arXiv:2503.13899.
7. T. Price-Broncucia\*, S. Amorese\*\*\*, R. Baptista, and **R. Morrison**. A probabilistic graphical model approach to interpret, verify, and decouple multi-physics systems. In *16th World Congress in Computational Mechanics (WCCM)*, 2024.
6. R. Bandy\*, T. Portone, and **R. Morrison**. Stochastic model correction for the adaptive vibration isolation round-robin challenge. In *Model Validation and Uncertainty Quantification (Proceedings of IMAC)*, pages 53–62. Springer Nature Switzerland, 2024.
5. T. Price-Broncucia\* and **R. Morrison**. Ultra-short time batching and unscented Kalman inversion for calibration of expensive chaotic models. *United States National Congress on Computation Mechanics*, Albuquerque, NM, 2023. (No proceedings recorded.) **UQ Student Paper Competition Semi-Finalist**.
4. T. Price-Broncucia\* and **R. Morrison**. Multi-time unscented Kalman inversion for calibration of expensive chaotic models. In *14th International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP14)*, Dublin, Ireland, 2023. (Acceptance rate: 45%) **CERRA Student Recognition Award**.
3. R. Bandy\* and **R. Morrison**. Quantifying model form uncertainty in spring-mass-damper systems. In *Model Validation and Uncertainty Quantification, Vol. 3 (Proceedings of the 41st IMAC)*, pages 9–19. Springer Nature Switzerland, 2023. **Best Paper Award**.
2. **R. Morrison**, R. Baptista, and Y. Marzouk. Beyond normality: Learning sparse probabilistic models in the non-Gaussian setting. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 11 pages, 2017. (Acceptance rate: 21%; journal equivalent)

1. **R. Morrison**, C. Bryant, G. Terejanu, K. Miki, S. Prudhomme. Optimal data split methodology for model validation. In *Proceedings of the World Congress on Engineering and Computer Science*, pages 1038-1043, 2011. (Acceptance rate: 53%) **Best Student Paper Award**.

## Theses

2. PhD Dissertation: *On the representation of model inadequacy: A stochastic operator approach*, Advisor: Prof. Robert Moser, UT Austin, 2016
1. Honors Senior Thesis: *From geometry to perturbation theory in combinatorial games: A case study of the game of Nim with a pass*, Advisor: Prof. Adam Landsberg, Scripps College, 2008

## Software

5. [Nonparanormal covariance \(examples\)](#). **R. Morrison**, June 2024.
4. [TransportMaps](#). D. Bigoni, A. Spantini, **R. Morrison**, R. Baptista, and Y. Marzouk. Transport Maps Software Documentation, 2015–2020.
3. [ARBO: Arbovirus Modeling and Uncertainty Quantification Toolbox](#). M. Tosin, E. Dantas, A. Cunha, and **R. Morrison**. ARBO Software Documentation, 2020–2024.
2. [Zika](#). **R. Morrison**. Initial release, 2020.
1. [Enriched-GLV](#). **R. Morrison**. Initial release, 2020.

## Presentations

### Colloquia & Invited Talks

- “Conditional and marginal independence after diagonal transport, and more generally.” Banff International Research Station, Workshop on Integrating Data- and Physics-Driven Methods for Decision Making under Uncertainty. Oaxaca, Mexico, 6/3/26
- “Analytic solutions to nonlinear ODEs via spectral power series.” Sandia National Laboratories, Center for Computing Research Seminar, CSRI. Albuquerque, NM, 2/23/26
- “Analytic solutions to nonlinear ODEs via spectral power series.” CU Boulder, Applied Math - Complex/Dynamical Systems Seminar. Boulder, CO, 2/12/26
- “Analytic solutions to nonlinear ODEs via spectral power series.” NIST, Applied and Computational Mathematics Division (ACMD) Seminar. Boulder, CO, 5/6/25
- “Analytic solutions to common nonlinear ODEs and why corresponding highly reduced models can work.” MD Anderson, Image-Driven Biology-Informed Therapy Program Seminar Series. Virtual, 4/29/24
- “Ensemble learning for accurate and reliable uncertainty quantification.” Space Weather Workshop. Boulder, CO, 4/18/24
- “Highly reduced models of random ODEs: A framework, some examples, and analysis.” IU Bloomington, IMAG (Interagency Modeling and Analysis Group)/ MSM (Multiscale Modeling) Viral Pandemics Seminar. Virtual, 4/4/24
- “Probabilistic graphical models: What they mean and how to find them.” Two Sigma Academic Seminar. New York City, NY, 3/14/24
- *Guest Lecture: Statistics and Data Science*. CU Boulder, Data Buffs (Student Chapter of the American Statistical Society) Meeting, Boulder, CO, 11/6/23
- “Structure, sparsity, & speed-ups: How sparse graphical models and approximate closed-form solutions can lead to exponential speed-ups for digital twins.” Santa Fe Institute, NSF Workshop: Crosscutting Needs for Digital Twins. Santa Fe, NM, 10/12/23
- “Structure and sparsity: From observations to data and back again.” CU Boulder, Computer Science Colloquium. Boulder, CO, 9/7/23

- “*Learning probabilistic graphical models with triangular transport and a Hessian score.*” Emory University, Mathematics Colloquium. Atlanta, GA, 8/31/23
- “*Triangular transport for learning probabilistic graphical models.*” Institute for Computational and Experimental Research in Mathematics (ICERM), Workshop on Optimal Transport in Data Science. Providence, RI, 5/11/23
- “*Learning probabilistic graphical models with triangular transport and a Hessian score.*” Colorado School of Mines, Applied Mathematics and Statistics Colloquium. Golden, CO, 5/5/23
- “*Learning sparse non-Gaussian graphical models.*” USACM TTA (Technical Thrust Area) on UQ and Probabilistic Modeling Seminar, **Inaugural talk of new seminar series**. Virtual, 2/10/21
- “*Learning non-Gaussian graphs.*” Rio de Janeiro State University, Computer Science Colloquium. Virtual, 8/25/21
- “*Learning sparse non-Gaussian graphical models.*” Caltech CMX (Computational Mathematics + X) Colloquium. Virtual, 12/9/20
- “*Learning sparse representations of model error in coupled differential equations.*” Potsdam Institute for Climate Impact Research, Workshop on Uncertainties in Data Analysis, **Keynote**. Virtual, 10/1/20
- “*Mathematical models in epidemiology.*” Computing Research Association (CRA) 2020 Virtual Conference Lightning Talk, **Contest winner** (Selected to appear on CRA website)
- “*Learning sparse non-Gaussian graphical models.*” Argonne National Laboratory, Laboratory for Applied Mathematics, Numerical Software, and Statistics (LANS) Seminar. Virtual, 5/20/20
- “*Representing model error in reduced models of interacting systems.*” CU Boulder, Applied Math Colloquium. Boulder, CO, 9/27/19
- “*Representing model inadequacy in reduced models of interacting systems.*” Statistical and Applied Mathematical Sciences Institute (SAMSI), Workshop on Statistical Perspectives on Uncertainty Quantification. Research Triangle Park, NC, 5/17/19
- “*Discovering dynamic interactions in complex systems.*” Johnson & Johnson Women in STEM2D Scholars Award Symposium. New Brunswick, NJ, 4/24/19
- “*Beyond normality: Learning sparse probabilistic graphical models in the non-Gaussian setting.*” Banff International Research Station, Workshop on Computational Uncertainty Quantification. Banff, Canada, 10/9/17
- “*Beyond normality: Learning sparse probabilistic graphical models in the non-Gaussian setting.*” Stanford, Institute for Computational Mathematics and Engineering, \*Talks Seminar. Palo Alto, CA, 5/1/17
- “*Towards reliable predictions: Bridging the gap between physics-based and data-driven models.*” CU Boulder, Computer Science Colloquium. Boulder, CO, 4/13/17
- “*Representing model inadequacy in reduced chemical mechanisms: A stochastic operator approach.*” MIT, Aerospace Computational Design Laboratory Seminar. Cambridge, MA, 2/26/16
- “*On the representation of model inadequacy: A stochastic operator approach.*” UT Austin, PhD Dissertation Defense. Austin, TX, 1/19/16
- Introduction to *The Theory That Would Not Die*. UT Austin, Babuška Forum Seminar Series at The Oden Institute. Austin, TX, 9/28/12
- “*Sensitivity of predicted quantities of interest for model validation.*” Sandia National Laboratories, Org. 1544 (Validation and UQ) Seminar. Albuquerque, NM, 8/15/12

## Conference Talks

- “*Conditional and marginal independence after diagonal transport, and more generally.*” Scientific Computing and Differential Equations (SciCADE). Edinburgh, UK, 7/3/26
- “*Analytic solutions to nonlinear ODEs via spectral power series and why highly reduced models can work.*” SIAM Conference on Uncertainty Quantification. Minneapolis, MN, 3/24/26

- “Explicit covariance formulas and estimation after diagonal transport.” SIAM Conference on Uncertainty Quantification. Trieste, Italy, 2/27/24
- “Highly reduced models of random ODEs: Some examples and analysis.” SIAM Conference on Mathematics of Data Science. San Diego, CA, 9/29/22
- “Learning graphs, density estimation, and the information gap.” SIAM Conference on Uncertainty Quantification. Virtual, 4/15/22
- “Representing model error in reduced models of interacting systems .” International Conference of the Engineering Mechanics Institute. Virtual, 3/24/21
- “Learning sparse dynamics of interacting systems.” 2nd Symposium on Machine Learning and Dynamical Systems, The Fields Institute. Virtual, 9/21/20
- “Representing model inadequacy in interacting systems.” SIAM Conference on Dynamical Systems. Snowbird, UT, 5/20/19
- “Representing model inadequacy in interacting systems with constrained stochastic operators.” SIAM Conference on Computational Science & Engineering. Spokane, WA, 3/01/19
- “Representing model inadequacy with physics-constrained data-driven stochastic operators.” World Congress on Computational Mechanics. New York City, NY, 7/24/18
- “Learning sparse non-Gaussian graphical models from data.” SIAM Conference on Uncertainty Quantification. Garden Grove, California, 4/18/18
- “Learning sparse probabilistic graphical models in the non-Gaussian setting.” SIAM Conference on Computational Science & Engineering. Atlanta, GA, 3/01/17
- “Representing model error in reduced chemical mechanisms: A stochastic operator approach.” Probabilistic Mechanics and Reliability Conference. Nashville, TN, 5/24/16
- “Assessing model error in reduced chemical mechanisms.” SIAM Conference on Uncertainty Quantification. Lausanne, Switzerland, 4/07/16
- “Representing model inadequacy in combustion mechanisms of laminar flames.” APS Division of Fluid Dynamics Annual Meeting. Boston, MA, 11/24/15
- “Stochastic inadequacy operators: A case study in methane combustion.” 13th US National Congress on Computational Mechanics. San Diego, CA, 7/27/15
- “Representing model inadequacy: A stochastic operator approach.” SIAM Workshop Celebrating Diversity. Salt Lake City, UT, 3/16/15
- “Stochastic inadequacy operators with applications to chemical kinetics.” SIAM Conference on Computational Science & Engineering. Salt Lake City, UT, 3/14/15
- “Representing model inadequacy in combustion kinetics.” APS Division of Fluid Dynamics Annual Meeting. San Francisco, CA, 11/25/14
- “Representing model inadequacy: Random matrix models with applications to chemical kinetics.” SIAM Conference on Uncertainty Quantification. Savannah, GA, 4/01/14
- “Data partition methodology for validation of predictive models.” SIAM Conference on Uncertainty Quantification. Raleigh, NC, 4/05/12
- “Optimal data split methodology for model validation.” The World Congress on Engineering and Computer Science 2011. Berkeley, CA, 10/20/11
- “Combinatorial games with a pass: A geometric approach.” SIAM Conference on Dynamical Systems. Snowbird, UT, 5/19/09

## Posters

- “Representing model inadequacy in reduced models of generalized Lotka-Volterra systems.” Reducing Dimensions and Cost for UQ in Complex Systems Workshop, Newton Institute. Cambridge, England, 3/5/18

- *"Beyond normality: Learning sparse probabilistic graphical models in the non-Gaussian setting."* Neural Information Processing Systems (NeurIPS). Long Beach, CA, 12/4/17
- *"Sensitivity of predicted quantities of interest for model validation."* Uncertainty Quantification Workshop, ICERM. Providence, RI, 10/10/12
- *"Data partition methodology for validation of predictive models."* Uncertainty Quantification Summer School, The University of Southern California. Los Angeles, CA, 8/23/12
- *"Sensitivity of predicted quantities of interest for model validation."* Predictive Science Academic Alliance Program (PSAAP) Meeting, University of Michigan. Ann Arbor, MI, 8/9/12
- *"Data partition methodology for validation of predictive models."* Large-Scale Inverse Problems and Quantification of Uncertainty Workshop, Institute for Mathematics and its Applications (IMA). Minneapolis, MN, 6/8/11
- *"Optimal data split methodology for model validation."* Uncertainty Quantification for Multiphysics and Multiscale Systems Workshop, The University of Southern California. Los Angeles, CA, 3/7/11

#### **Other Workshops, etc. — Attended but did not present**

- 18th US National Congress on Computational Mechanics, Chicago, IL, 7/(20–24)/25
- Uncertainty Quantification Workshop, Oberwolfach (Mathematisches Forschungsinstitut Oberwolfach), Germany, 4/(20–25)/25
- Rigorous and Reproducible Scientific Reasoning (R2SR) NSF Workshop, CU Boulder, CO, 11/(13–14)/23
- 50th Anniversary Workshop, The Oden Institute for Computational Engineering & Sciences, UT Austin, 9/21/23
- US National Congress on Computational Mechanics, Albuquerque, NM, 7/(23–27)/23
- Scientific Machine Learning Workshop, Banff International Research Station, Banff, Canada, 6/(19–23)/23
- Space Weather with Quantification of Uncertainty Workshop, MIT, Cambridge, MA, 3/(9–10)/23
- Climate Informatics Workshop, National Center for Atmospheric Research, Boulder, CO, 9/(20–21)/18
- MIT Statistics and Data Science Center Day (SDSCon 2017), Cambridge, MA, 4/21/17
- Uncertainty Quantification and Data-driven Modeling Workshop, Sandia/The Oden Institute, UT Austin, TX, 3/(23–24)/17
- Integrated Analysis for Agricultural Management Strategies Workshop, American Institute of Mathematics (AIM), San Jose, CA, 5/(4–8)/15
- Large-Scale Inverse Problems and Quantification of Uncertainty: Big Data Meets Big Models Workshop, Santa Fe, NM, 5/(22–24)/13
- Association for Women in Mathematics (AWM) Research Symposium, Santa Clara University, CA, 3/(16–17)/13
- Uncertainty Quantification School taught by Tony O'Hagan, UT Austin, TX, 10/(3–4)/11
- SAMSI/Sandia Summer School on Uncertainty Quantification, Albuquerque, NM, 6/(20–24)/11
- Women in Theory Workshop, Princeton, NJ, 6/(19–23)/10

#### **Student & Postdoc Advising - My Research Group**

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##### **Postdoc**

- Jack Shaw (joint position with Geological Sciences), 2024 – present

##### **PhD**

- Robert Dumitrescu, joining Fall 2026
- Mahi Khoshnevisan, joining Fall 2026
- Nate Holland, 2025 – present

- Ujas Shah, 2024 – present
  - Completed preliminary exam 6/11/26
- Teo Price-Broncucia, 2019 – 2025
  - Thesis: *From Ages to Instants: Calibration and Error Detection for Expensive Chaotic Models using Short Runs, Ensembles, and Approximate Gradients* (defended 2025)
  - Next/current position: Advanced Study Program (ASP) Postdoctoral Fellow, NCAR
  - Awards: USNCCM23 UQ Student Paper Competition Semi-Finalist; ICASP14 2023 CERRA Student Recognition Award; 2024 Caelan B. Lapointe Memorial Graduate Scholarship (2024); selected for the Rising Stars in Computational & Data Sciences Workshop (2025)
- Rileigh Bandy, 2019 – 2024
  - Thesis: *Uncertainty Representations in White- and Black-Box Models: Quantifying Model-Form and Measurement Errors in Computational Science* (defended 5/8/24)
  - Next position: Postdoc, Sandia National Laboratories; currently BAE Systems
  - Awards: IMAC23 Best Paper Award; selected for the Rising Stars in Computational & Data Sciences Workshop (2024)

### Master's

- Rachel Washington (summer intern, non-thesis student), 2024

### Undergraduate

- Josiah Shehata, CS Senior Thesis, joining Fall 2026
- Sienna Amorese (Applied Math: Stats), Latin Honors Senior Thesis, 2024 – 2025
- Daniel Crook, Discovery Learning Apprenticeship (DLA), 2020 – 2021
- Michael Donovan, DLA, 2020 – 2021

### Other Student Advising

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Only the most recent exam is listed when multiple. Totals: 15 PhD dissertation/qualifying committees, 1 Master's thesis, 1 senior thesis.

### PhD Thesis Committee Member

- Leila Ghaffari (CS) 06/26/25 (Second reader)
- Ren Stengel (CS) 06/23/25 (Second reader)
- Mike McCabe (CS) 4/2/24
- Masu Parmar (AERO) 7/17/23
- Jingwei Li (CS) 4/10/23
- Chou Yi (CS) 7/9/21
- Sebastian Laudenschlager (CS) 5/27/21
- Felix Newberry (AERO) 4/2/21
- Zhiyuan Liu (CS) 4/1/21
- Tristan Konolige (CS) 4/2/20
- Paul Diaz (AERO) 2/25/20
- Eric Peters (AERO) 5/3/19

### PhD Area/Comprehensive Exam Committee Member

- Ellery Galvin (APPM) 4/22/26
- Nicholas Rummel (APPM) 3/10/26



- Sarah Gage (CS) 11/8/19

#### Master's Thesis Committee Member

- Caleb Kumar (CS) 4/7/26

#### Senior Thesis Committee Member

- Saurabh Totey (PHYS) 5/4/23

#### Courses Taught & Developed

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Course number and title; times taught; level (Newly developed courses noted.)

- CSCI 2820 *Linear Algebra with Computer Science Applications* — 3 — Undergraduate
- CSCI 4802/5802 *Data Science Team* — 2 — Mixed
- CSCI 5646 *Numerical Linear Algebra* — 2 — Graduate
- CSCI 5822 *Probabilistic and Causal Modeling in Computer Science* — 2 — Graduate
- CSCI 6110 *Graduate Writing Workshop (developed)* — 3 — Graduate/Mixed
- CSCI 6166 *Validation and Uncertainty Quantification for Computational Models (developed)* — 3 — Graduate
- CSCI 7000 *Graduate Writing Workshop, Part 2 (developed)* — 1 — Graduate/Mixed
- CSCI 7000 *Research & Reading Group UPSCALE (Uncertainty, Probability, Scientific Computing And LEarning) (developed)* — 1 as formal class, 11 as informal group — Graduate
- CSCI 4900/APPM 6900 *Independent Study* — 6 — Graduate

#### By Term — Title (Enrollment); Notes

*Spring 26*

- CSCI 7000 *Graduate Writing Workshop, Part 2* (12); New course to build writing curriculum in STEM

*Fall 25*

- CSCI 5646 *Numerical Linear Algebra* (10)
- CSCI 6110 *Graduate Writing Workshop* (23)

*Spring 25*

- CSCI 6166 *Validation and Uncertainty Quantification for Computational Models* (9)

*Fall 24* — On leave

*Spring 24*

- CSCI 7000 *Research & Reading Group UPSCALE* (13); New course about state-of-the-art research
- APPM 6900 *Independent Study* (2)

*Fall 23*

- CSCI 5646 *Numerical Linear Algebra* (35)
- CSCI 6110 *Graduate Writing Workshop* (13)
- APPM 6900 *Independent Study* (1)

*Spring 23*

- CSCI 5822 *Probabilistic and Causal Modeling in Computer Science* (26)

*Fall 22*

- CSCI 2820 *Linear Algebra with Computer Science Applications* (78)

*Spring 22* — On leave

*Fall 21*

- CSCI 6110 *Graduate Writing Workshop* (24); New course to build writing curriculum in STEM

#### *Spring 21*

- CSCI 5822 *Probabilistic and Causal Modeling in Computer Science* (42); Redesigned with emphasis on graphical models
- CSCI 4802/5802 *Data Science Team* (41)
- CSCI 4900 *Independent Study* (1)

#### *Fall 20*

- CSCI 4802/5802 *Data Science Team* (20)

#### *Spring 20*

- CSCI 6166 *Validation and Uncertainty Quantification for Computational Models* (5)

#### *Fall 19*

- CSCI 2820 *Linear Algebra with Computer Science Applications* (77)
- CSCI 4900 *Independent Study* (1)

#### *Spring 19*

- CSCI 2820 *Linear Algebra with Computer Science Applications* (73); Redesigned (QR factorization, clustering, least squares, applications)

#### *Fall 18*

- CSCI 6166 *Validation and Uncertainty Quantification for Computational Models* (3); New course based on my research area and notes from R. Moser
- CSCI 4900 *Independent Study* (1)

#### *Fall 19 – present*

- Research & Reading Group UPSCALE (About 10–15 students & postdocs); Faculty Founder and Sponsor

## Teaching & Professional Development

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- Graduate Inclusive Mentoring Community (GIMC) Session: Communication and Conflict Resolution Strategies for Success, 2/19/26
- GIMC Happy Hour, Norlin Library, CU Boulder, 11/20/25
- Fall Teaching, Learning, and Technology Conference 2025 - Engaging Every Learner, Center for Teaching & Learning (CTL), CU Boulder, 10/2/25
- Teaching Circles Participant, CS Department, CU Boulder, Fall 2025
- Teaching Circles Participant, CS Department, CU Boulder, Spring 2025
- TQF Hackathon, CS Department, CU Boulder, 04/14/24
- Teaching Circles Participant, CS Department, CU Boulder, Spring 2023
- Fall 2020 Diversity and Inclusion Summit, CU Boulder, 11/10/20
- “What’s Your Plan? A Symposium on Education for Fall 2020,” College of Engineering and Applied Science, CU Boulder, 7/27/20
- 2020 Career Mentoring Workshop, Computing Research Association (CRA), Washington DC, 2/(27–28)/20
- Introductory Leadership Training Workshop, Leadership Education for Advancement & Promotion (LEAP), CU Boulder, 1/(8–9)/20
- New Assistant Professor Program Certificate, Faculty Teaching Excellence Program (FTEP), CU Boulder, 2018 – 2019

- Kaufman Teaching Certificate Program (Intensive 2-Week Course), MIT, 5/29 – 6/15/18
- NSF CISE CAREER Proposal Writing Workshop, Washington, DC, 4/(9–10)/18

## Professional Service

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### *Society & Conference Leadership*

- **Co-Chair**, SIAM Conference on Mathematics of Data Science (MDS26), Salt Lake City, 2026 — co-leading the scientific program for the field's flagship meeting (~1,000 attendees) over an 18-month organizing cycle: plenary selection, minisymposia review, and program assembly.
- **SIAM Liaison** to the Joint Mathematics Meetings (JMM), 2024 – 2026 (**Chair**, 2025) — three-year term coordinating SIAM's programming at the AMS flagship meeting.
- **Program Director**, SIAM Activity Group on Data Science, 2024 – 2025 — **co-founded WebMDS** (Webinar on Mathematics of Data Science), a public webinar series, and moderated two panels (on teaching, and on careers, in the mathematics of data science); through this role, organized a Data Science track as part of the SIAM Annual Meeting 2025 Organizing Committee.

Moderator, [WebMDS Panel: Careers in Mathematics of Data Science](#), 8/12/25

Moderator, [WebMDS Panel: Teaching the Mathematics of Data Science](#), 5/6/25

- Member, SIAM UQ Early Career Prize Committee, 2025
- Secretary, SIAM Activity Group on Uncertainty Quantification, 2021 – 2022
- Co-Organizer, minisymposia at SIAM UQ (2022, 2024), USNCCM (2021), and ECCOMAS (2022)
- Organizer, Workshop Celebrating Diversity at SIAM Annual 2020 and SIAM CSE 2019

### *Review & Panel Service*

- **NSF review panels:** CISE Directorate (2024) and MPS/DMS Directorate (2026); NSF reviewer (2023, 2024) — each panel entails reading and reviewing 10–15 proposals
- **Manuscript refereeing:** ~[6–10] reviews per year for 20+ journals, conferences, and presses, including: *Advances in Neural Information Processing Systems (NeurIPS)*, *Chaos*, *Computational Science and Engineering*, *Computer Methods in Applied Mechanics and Engineering*, *Discrete Applied Mathematics*, *Entropy*, *Games of No Chance*, *IEEE Computing in Science and Engineering*, *IEEE/ACM Transactions on Networking*, *International Conference on Machine Learning (ICML)*, *International Journal of Uncertainty Quantification*, *Journal of Physics: Complexity*, *Machine Learning: Science and Technology*, *Mathematical Reviews*, *Mathematics*, *The MIT Press*, *Nature Computational Science*, *Nonlinear Dynamics*, *Pattern Recognition Letters*, *Physica D*, *SIAM Journal on Scientific Computing*, *SIAM/ASA Journal on Uncertainty Quantification*, *Space Weather*
- **Awards refereeing:** Reviewer, NCWIT Award for Aspirations in Computing (four years: 2020, 2021, 2023, 2024; e.g., 10 applications in 2023); Poster Judge, SIAM Mathematics of Data Science, 2022

### *Mentoring & Broadening Participation*

- Faculty Mentor, USACM Computational Mechanics Student Mentorship Program (CMSMP) — paired for a year with graduate students at other institutions (Stanford, Chapman), 2024 – 2026
- Mentor, Johnson & Johnson Women in STEM2D Program, 2024 – 2025
- “Lunch Mentor,” USNCCM17 (Albuquerque, 2023) and USNCCM18 (Chicago, 2025) Mentoring Events and “Breakfast Mentor,” Rocky Mountain Celebration of Women in Computing, 2018
- Faculty Mentor, Peak to Peak Charter School (Lafayette, CO), 2023
- Volunteer Tutor for 8th-grade math, Austin Partners in Education (APIE), 2013 – 2014

## University Service

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### *Governance & Community*

- **CS Executive Committee Member**, 08/2023 – 05/2025 — the department's senior governance committee. Created and led **CS++**, a monthly faculty/staff social hour, and helped institute monthly all-department meetings with standing-committee reports.
- Helped draft CU Boulder's **Computational Science and Engineering (CSE) certificate**.
- **CS Faculty Search Committee Member**, 2022 – 2023 and 2023 – 2024 (two cycles); ECEE Faculty Search Committee Member, 2020 – 2021
- CS Undergraduate Committee Member, 08/2025 – present; CS Graduate Curriculum Committee Member, 01/2020 – 12/2021; CS Undergraduate Curriculum Committee Member, 08/2018 – 12/2019
- Representative for the CS Junior Faculty Assembly, 2022; School of Computing Task Force Member, Spring 2019; Covid-19 CEAS Working Group Member, Summer 2020

#### *Outreach, Mentoring, & Internal Review*

- CEAS NSF GRFP Introductory Workshop (presenter), 2023 and 2024 — coached ~30 graduate students on fellowship applications
- External Funding Opportunities Workshop (presenter), 8/14/25
- Panel Member, LEAP Workshop Session “Mentoring Students: Ask the Stars,” 05/08/26
- Faculty Mentor, BOLD Center RedShirt S-STEM Program, 2019 – 2020
- Grant reviewer, CU Research & Innovation Office (RIO), 2019 and 2020; AB Nexus Reviewer, 2023; reviewer, Johnson & Johnson WiSTEM2D Award (internal review), 2020 and 2021

#### *Before August 2018*

- Organizer for Babuška Forum Series (UT Austin), with Professor Ivo Babuška, 2013 – 2014
- Graduate Student Member, Student Health Advisory Committee (SHAC), UT Austin, 2012 – 2013
- Representative from the Oden Institute to the Graduate Student Assembly (GSA), UT Austin, 2011 – 2013
- Graduate Student Representative for Computational Science, Engineering, and Mathematics (CSEM), 2010 – 2012

#### Outreach & Media

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- [15 Faculty Recognized as Exceptional Mentors](#) CU Graduate School, 5/2/25
- [How did the division of labor emerge in animals and humans? Little-known organisms hold clues](#) CU Boulder Today, 11/12/24
- [Group profile by Grace Wilson](#) CU Boulder, CS Department 5/4/24
- [See how WiSTEM2D Scholars Award Winner Rebecca Morrison is using flexible algorithms to address complex climate issues](#) Johnson & Johnson, 08/05/22
- [Interview with Jessie Finocchiaro](#) CU Boulder CS Neural Network, 4/23/21
- [Covid models show how to avoid future lockdowns](#) Scientific American, 11/24/20
- [Covid-19: Why most virus models are wrong](#) Westword, 05/12/20
- [Scientists develop tool to improve disease model accuracy](#) CU Boulder Today, 05/05/20
- [Assistant professor wins research award from Johnson & Johnson](#) CU Boulder Today, 04/12/19

#### Professional Memberships

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- American Mathematical Society (AMS)
- Association for Women in Mathematics (AWM)
- Society for Industrial and Applied Mathematics (SIAM)
- United States Association for Computational Mechanics (USACM)